

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response
Questions

19 September 2022

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question on the separate Answer Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

Section A

Answer **all** the questions in this section.

- 1 Human lactase is a transmembrane protein located in the brush border of the human small intestines. In humans, the lactase gene, *Lct*, is found on chromosome 2. Transcription and translation result in the formation of pre-pro-lactase (1927 amino acids), which is cleaved (cut) into pro-lactase (1023 amino acids) and then converted into its mature form, lactase.

Fig 1.1 shows the schematic diagram of processing and localization of the human lactase translational product.

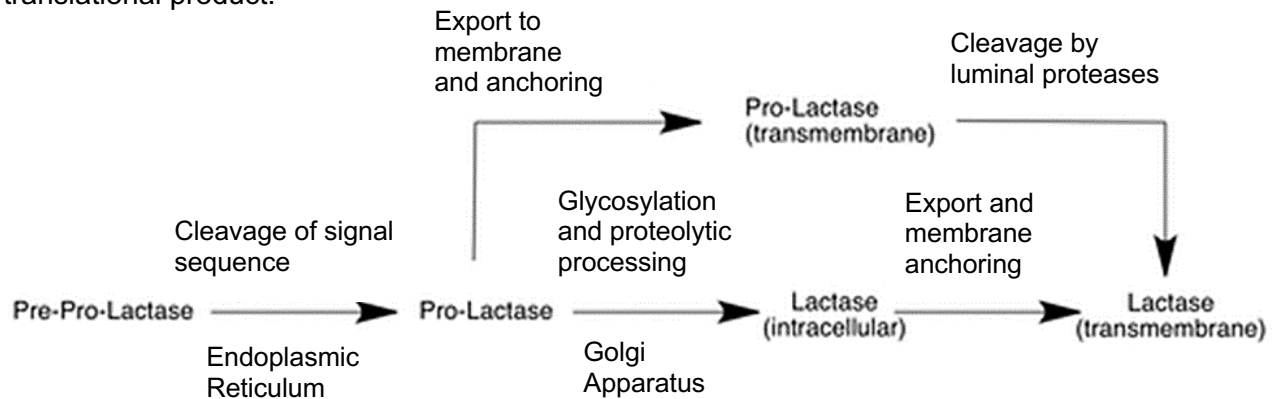


Fig. 1.1

- (a) The primary translational product of the human *Lct* and rat lactase gene differ by only a single amino acid.
Suggest reasons for the similarity in the structure of human *Lct* and rat *Lct*.

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- (b) The pre-pro-lactase molecule is cleaved in the rough endoplasmic reticulum.
With reference to Fig. 1.1, describe the sequence of events occurring between cleavage in the rough endoplasmic reticulum and export of the molecule.

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- (c) Suggest why the signal sequence in pre-pro-lactase needs to be cleaved to form pro-lactase.

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Research has shown that newborns express high levels of lactase. However, in most of the world's population, lactase transcription is down-regulated after weaning. Weaning is the process by which babies who were fully reliant on milk are introduced to solid foods.

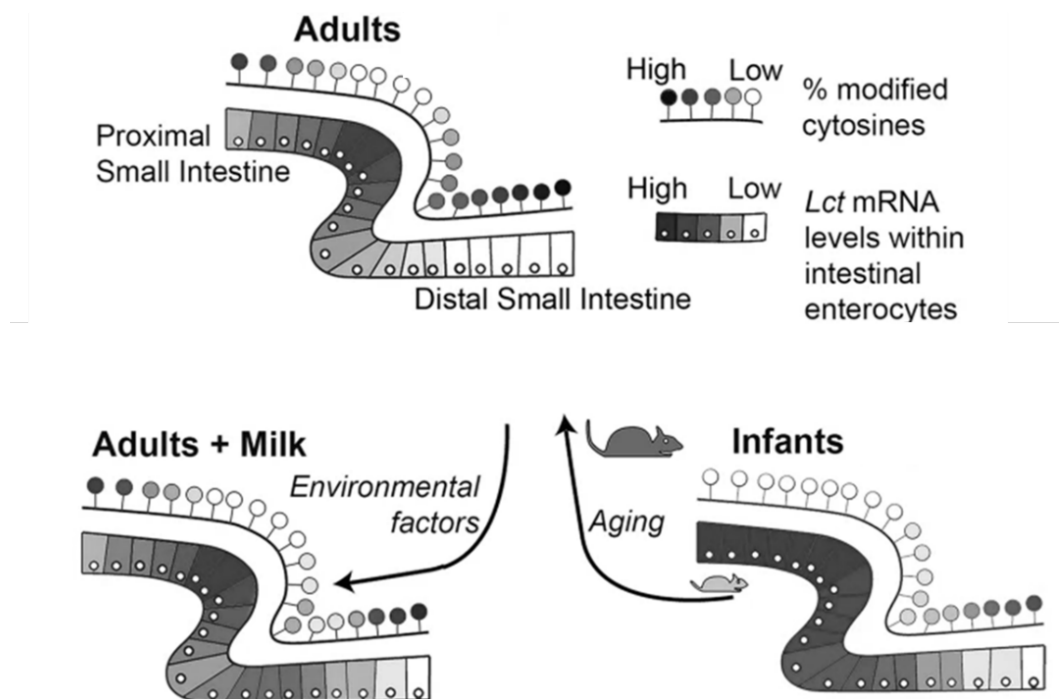


Fig. 1.2

- (d) With reference to Fig. 1.2,
- (i) Describe how the expression of *Lct* is influenced by environmental factors and by age.

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(ii) Explain how the down-regulation of lactase transcription occurred.

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Lactose is the primary disaccharide in almost all mammalian milk. Cow and goat milk are some of the most common milk that is widely consumed all over the world. Some people become lactose-intolerant because they cease to produce enough of the enzyme lactase. Lactase digests lactose to glucose and galactose.

(e) Complete Fig. 1.3 to show how lactose is broken down into glucose and galactose by the enzyme lactase.

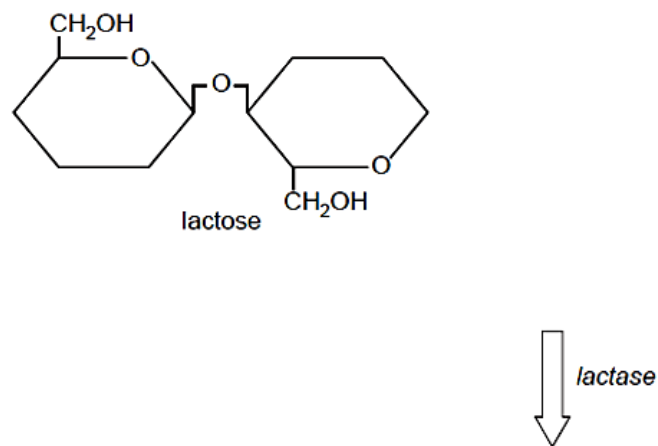


Fig. 1.3

[3]

- (f) Lactose-intolerant people suffer from severe discomfort, including diarrhoea, when they consume cow's milk or eat dairy products.

Fig. 1.4 shows the results of an investigation comparing:

- the effects on 50 lactose-intolerant volunteers of drinking normal cow's milk
- the effects on the same 50 lactose-intolerant volunteers of drinking low lactose cow's milk
- the effects on a control group of 15 volunteers, who were not lactose-intolerant of drinking normal cow's milk.

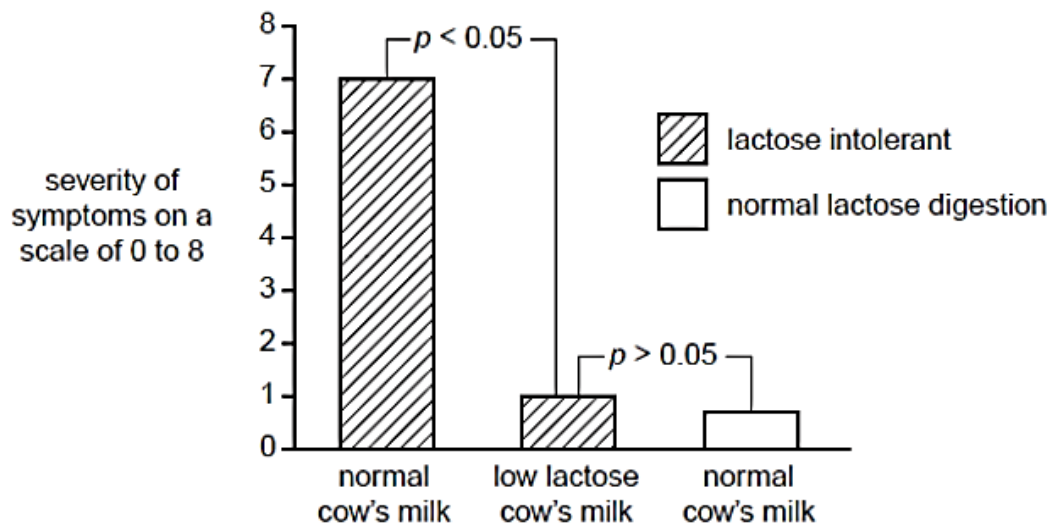


Fig. 1.4

With reference to the probability values shown in Fig. 1.4, comment on the results of this investigation.

$P < 0.05$:

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$P > 0.05$:

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- (g) Immobilised lactase can be used commercially to produce low lactose cow's milk from normal cow's milk for people who are lactose intolerant.

Discuss the advantages of using immobilised lactase rather than lactase that is not immobilised.

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- (h) Due to an overlap with the symptoms, such as diarrhoea and vomiting, milk allergy is often confused with lactose intolerance. A food allergy happens when the immune system of a person overreacts to a specific food protein and it is associated with abnormally elevated antibody levels.

Describe how the presence of cow milk proteins induces the production of antibodies by B lymphocytes.

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[Total: 25]

- 2 Over thousands and millions of years, there have been natural cycles in the Earth's climate. There have been ice ages (e.g. the Quaternary Ice Age) and warmer interglacial periods. Climatic changes can be investigated using evidence left in tree rings, layers of ice in glaciers, ocean sediments and layers of sedimentary rocks. For example, bubbles of air in glacial ice trap tiny samples of Earth's atmosphere, giving scientists a history of greenhouse gases that stretches back more than 800,000 years.

These changes in climate affect plant distribution and physiology. Fig. 2.1 is a diagram showing the profile of two mountains in the tropics during a warm phase and a cool phase in the Earth's climate. The distribution of rain forest vegetation is also shown.

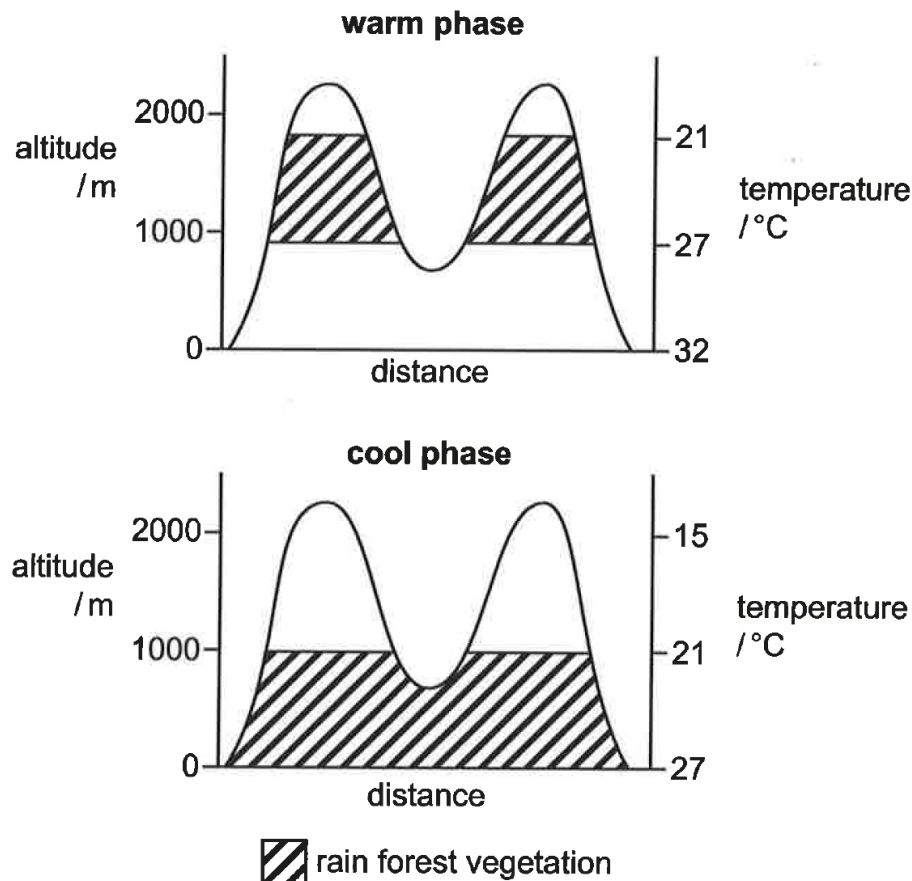


Fig. 2.1

- (a) (i) Describe and explain the effect of climate change on the distribution of rain forest vegetation in the tropics, as shown in Fig. 2.1.

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- (ii) Over millions of years there are repetitive cycles of alternating warm and cool phases in the Earth's climate.

Suggest how repeated changes in climate between the two phases shown in Fig. 2.1 may lead to greater species diversity.

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Changes in climate also affect insect physiology. Dengue virus is endemic in Singapore and the dengue virus vector is *Aedes aegypti*.

To study the influence of weather conditions on the incidence of dengue viral infection in Singapore, scientists recorded the amount of rainfall, mean weekly temperatures and the number of dengue cases over a one-year period. The recorded data is shown in Fig. 2.2.

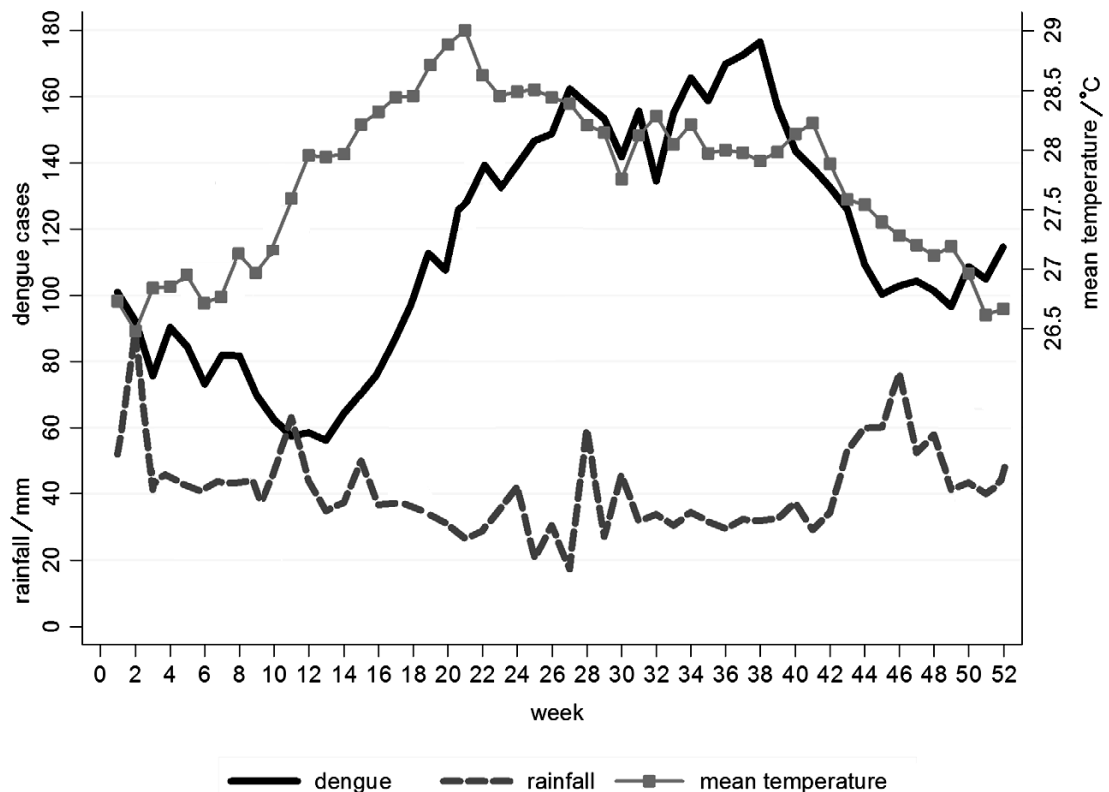


Fig. 2.2

- (b) (i)** Based on the information provided, comment on which weather condition is a better predictor of dengue cases in Singapore.

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- (ii)** Suggest explanations for the relationship between the weather condition chosen in part **(i)** and the number of dengue cases recorded.

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[Total: 12]

- 3 Apoptosis is an important part of tissue homeostasis. It ensures that cells die quickly and are safely removed.

Every cell undergoing apoptosis shows a predictable series of changes. The cell shrinks and breaks up into dead cell fragments bounded by membranes. The membranes of these dead cell fragments display unique surface ligands, which identify the fragments to neighbouring cells ingesting these fragments.

Before the apoptotic cell breaks up, cellular DNA is degraded by endonucleases into small pieces that each differ in size by 180 base pairs.

Fig. 3.1 is an electron micrograph of a cell with ingested dead cell fragments.

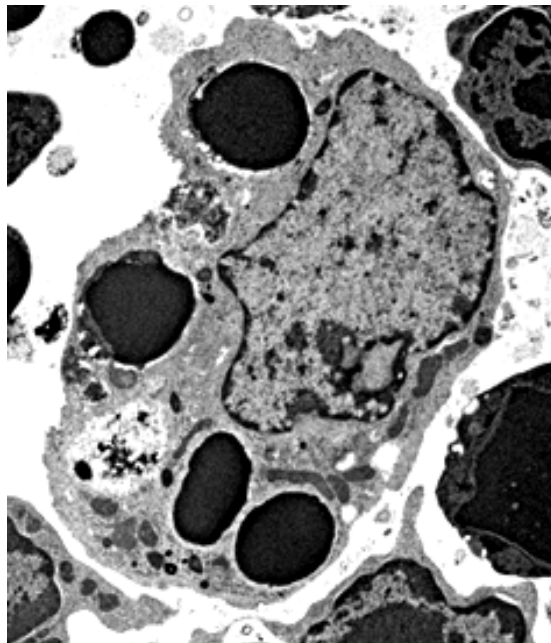


Fig. 3.1

- (a) (i) Name **one** type of cell that can ingest dead cell fragments and explain how the cell recognises these cell fragments.

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- (ii) Suggest how the DNA isolated from apoptotic cells can be analysed to confirm that apoptosis had indeed been activated.

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(b) Fig. 3.2 shows one way in which apoptosis can be triggered.

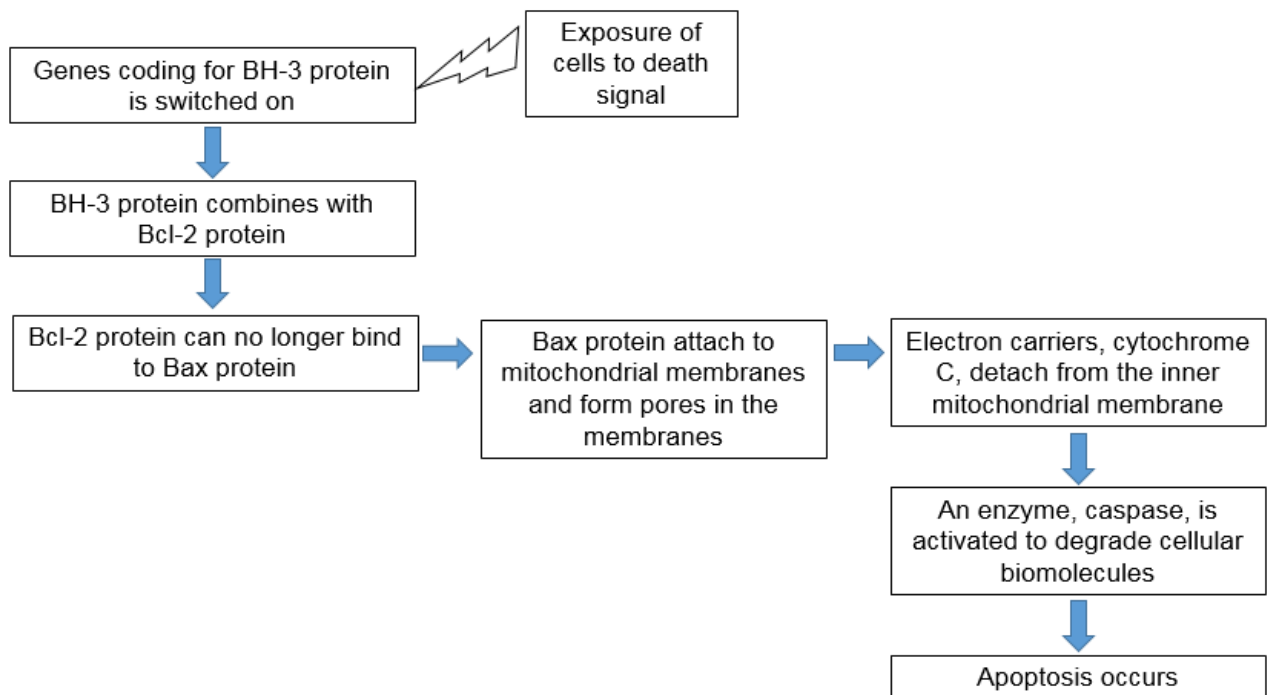


Fig. 3.2

- (i) With reference to Fig 3.2, predict and explain how ATP production will change when cytochrome C is detached from the inner mitochondrial membrane.

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- (ii) A particular mutation leads to the excessive production of Bcl-2 protein, disrupting the apoptotic pathway, leading to B-cell lymphoma, a cancer of B-cells.

- Bcl-2 gene is found on chromosome 18.
- In B-cell lymphoma, a mutation shifts the Bcl-2 gene to chromosome 14, bringing it close to a powerful gene regulatory sequence.
- This regulatory sequence regulates the production of the immunoglobulin heavy chain.

Explain why the mutation will lead to B-cell lymphoma.

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[Total: 13]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a)** Describe how genetic stability in eukaryotes is maintained at the molecular, cellular and population levels. [15]

- (b)** Discuss the importance of water in living organisms. [10]

[Total: 25]

- 5 (a)** Cells are the smallest unit of life, and all their activities (processes and functions) are tightly controlled. [15]

Describe how cells control their activities and explain the importance of these controls.

- (b)** Discuss the importance of nitrogen in living organisms. [10]

[Total: 25]